1. Find the sample space associated with the experiment of rolling a pair of dice (one is blue and the other red) once. Also, find the number of elements of this sample space.
2. In each of the following experiments specify appropriate sample space.
    a) A boy has a 1 rupee coin, a 2 rupee coin and a 5 rupee coin in his pocket. He takes out two coins out of his pocket, one after the other.
    b) A person is noting down the number of accidents along a busy highway during a year.
3. A coin is tossed. If it shows head, we draw a ball from a bag consisting of 3 blue and 4 white balls; if it shows tail we throw a die. Describe the sample space of this experiment.
4. An experiment involves rolling a pair of dice and recording the numbers that come up. Describe the following events:
   A: the sum is greater than 8
   B: 2 occurs on either die
   C: the sum is at least 7, and a multiple of 3.
   Which pairs of these events are mutually exclusive?
5. A bag contains 9 discs of which 4 are red, 3 are blue and 2 are yellow. The discs are similar in shape and size. A disc is drawn at random from the bag. Calculate the probability that it will be (i) red, (ii) yellow, (iii) blue, (iv) not blue, (v) either red or blue.
6. Find the probability that when a hand of 7 cards is drawn from a well shuffled deck of 52 cards, it contains (i) all Kings (ii) 3 Kings (iii) at least 3 Kings.
7. A family has two children. What is the probability that both the children are boys given that at least one of them is a boy?
8. An urn contains 10 black and 5 white balls. Two balls are drawn from the urn one after the other without replacement. What is the probability that both drawn balls are black?
9. Two cards are drawn successively with replacement from a well-shuffled deck of 52 cards. Find the probability distribution of the number of aces.
10. Find the probability distribution of number of doublets in three throws of a pair of dice.
11. Certainly! Here are 10 situations. For each one, determine whether the random variable involved is discrete or continuous:
    a) The number of cars passing through a toll booth in one hour.
    b) The height of a randomly selected person.
    c) The number of goals scored in a soccer match.
    d) The amount of milk (in liters) a cow produces each day.
    e) The number of students in a classroom.
    f) The temperature in a city at noon.
    g) The number of emails received in a day.
    h) The weight of a randomly chosen package from a shipping company.
    i) The time it takes for a light bulb to burn out (in hours).
    j) The number of pages in a book.

12. A random variable has the following probability distribution :

X	:	0	1	2	3	4	5	6	7	8
P(X=x)	:	k	3k	5k	7k	9k	11k	13k	15k	17k

(a) Determine the value of k.
(b) Find p[X<4], p[0<x<4].
(c) Find the distribution function.
(d) Find the smallest value for x for which p[X≤x]>1/2.

13. The p.m.f of a random variable X is zero except at the points x= 0,1,2. At these points, it has the values $p(0)=3c^3$, $p(1)=4c-10c^3$, and $p(2)=5c-1$, for some c>0.
(a) Find the value c.
(b) Evaluate p(X<2), and p(1<X≤2).
(c) Distribution function.

14. The Diameter of an electric cable, say X, is assumed to be a continuous random variable with p.d.f f(x)= 6x(1-x), 0≤x≤1.
(a) Check that the above is p.d.f.
(b) Determine a number 'b' such that p(X<b)=p(X>b)

15. A box contains 12 bulbs of which 3 are defective. If 3 balls are drawn from the box at random, find the probability distribution of X, the number of defective bulbs drawn.

16. verify that the following is a distribution function:

$$F(x)=\begin{cases}0, & x<0\\ \frac{1}{2}\left(\frac{x}{a}+1\right), & -a\le x\le a\\ 1, & x>1\end{cases}$$

17. Let X be a continuous random variate with p.d.f.

$$f(x)=\begin{cases}ax, & 0\le x\le 1\\ a, & 1\le x\le 2\\ -ax+3a, & 2\le x\le 3\\ 0, & elsewhere\end{cases}$$

(i) Determine the constant a.
(ii) Compute p(X≤1.5).

18. Let p (x) be the probability function of a discrete random variable X which assumes the values x_1, x_2, x_3, x_4, such that 2 p (x_1) = 3p (x_2) = p (x_3) = 5 p (x_4). Find the probability distribution and cumulative probability distribution of X.

19. Let X be a random variable such that
$P(X=-2)=P(X=-1)$, $P(X=2)=P(X=1)$ and
$P(X>0)=P(X<0)=P(X=0)$.
Obtain the probability mass function of *X* and its distribution function.

20. Consider the experiment of tossing two dice. Let X denote the total of the upturned faces. Find the probability function of X. If Y denotes the absolute difference of the upturned faces, Find the probability function of Y.

21. Consider the experiment of tossing three coins. Let X denote the number of heads. Find the probability function of X. Also find the cumulative distribution function F(x).

22. A random variable X has the p.d.f

$$f(x)=\begin{cases}2x, & 0<x<1\\ 0, & \textit{otherwise}\end{cases}$$

Find $P(X<1), P\left(\frac{1}{4}<X<\frac{1}{2}\right)$